

SECTION B

Question 04

```
import matplotlib.pyplot as plt
from collections import Counter
import re
```

```
def visualized_word_freq (filename):

    # 1. Read the text from the file
    with open(filename, 'r') as file:
        text_content = file.read()

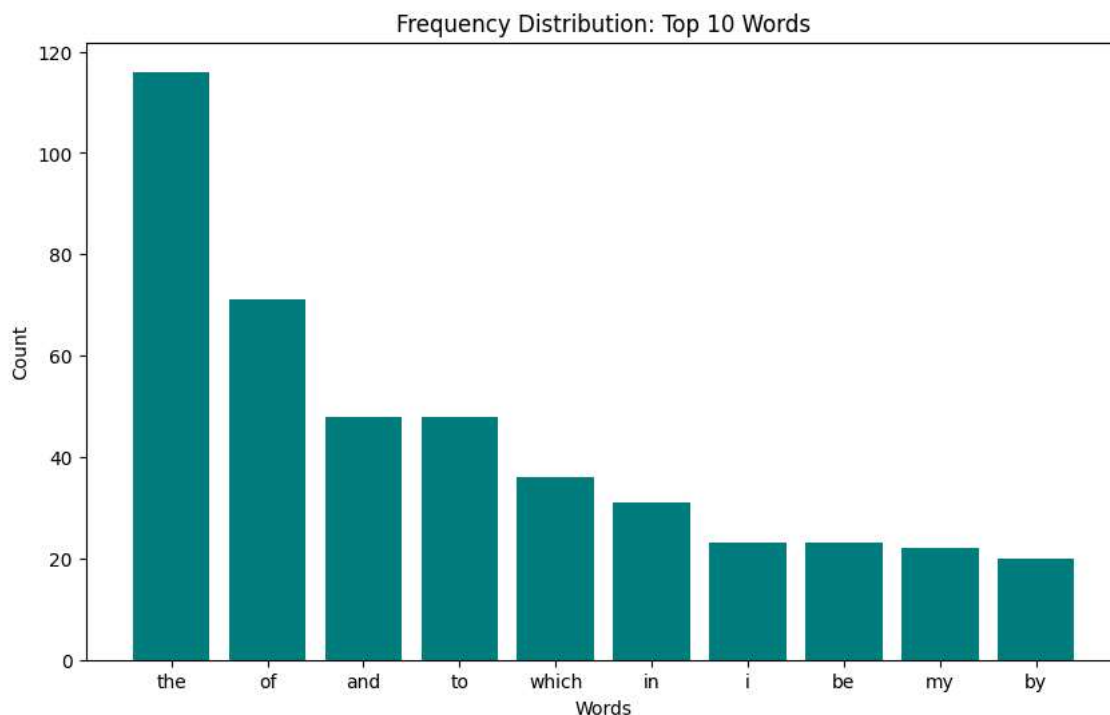
    # Clean the text (lowercase and remove non-alphanumeric noise)
    words = re.findall(r'\w+', text_content.lower())

    #2 Frequency Distribution
    freq_dist = Counter(words)
    top_10 = freq_dist.most_common(10)

    #3 Visualize
    words_labels, counts = zip(*top_10)

    # 4. Create the bar chart
    plt.figure(figsize=(10, 6))
    plt.bar(words_labels, counts, color='teal')
    plt.title('Frequency Distribution: Top 10 Words')
    plt.xlabel('Words')
    plt.ylabel('Count')
    plt.show()

# Call the function with the desired filename
visualized_word_freq(r"/content/1789-Washington.txt")
```



Question 05

```
import numpy as np
import matplotlib.pyplot as plt

def calculate_bce_loss(y_true, y_pred):
    epsilon = 1e-15
```

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y_pred = np.clip(y_pred, epsilon, 1 - epsilon)
loss = -np.mean(y_true * np.log(y_pred) + (1 - y_true) * np.log(1 - y_pred))
return loss

# Simulate Training over 10 Epochs
epochs = list(range(1, 11))
# Simulated loss values showing a typical learning curve (decreasing over time)
simulated_loss = [0.82, 0.65, 0.51, 0.42, 0.35, 0.28, 0.22, 0.18, 0.15, 0.12]

# Visualization
plt.figure(figsize=(8, 5))
plt.plot(epochs, simulated_loss, marker='o', color='darkorange', label='BCE Loss')
plt.title('Model Training Progress (Loss Tracking)')
plt.xlabel('Epochs')
plt.ylabel('Loss Value')
plt.grid(True, linestyle='--')
plt.legend()
plt.show()
```

